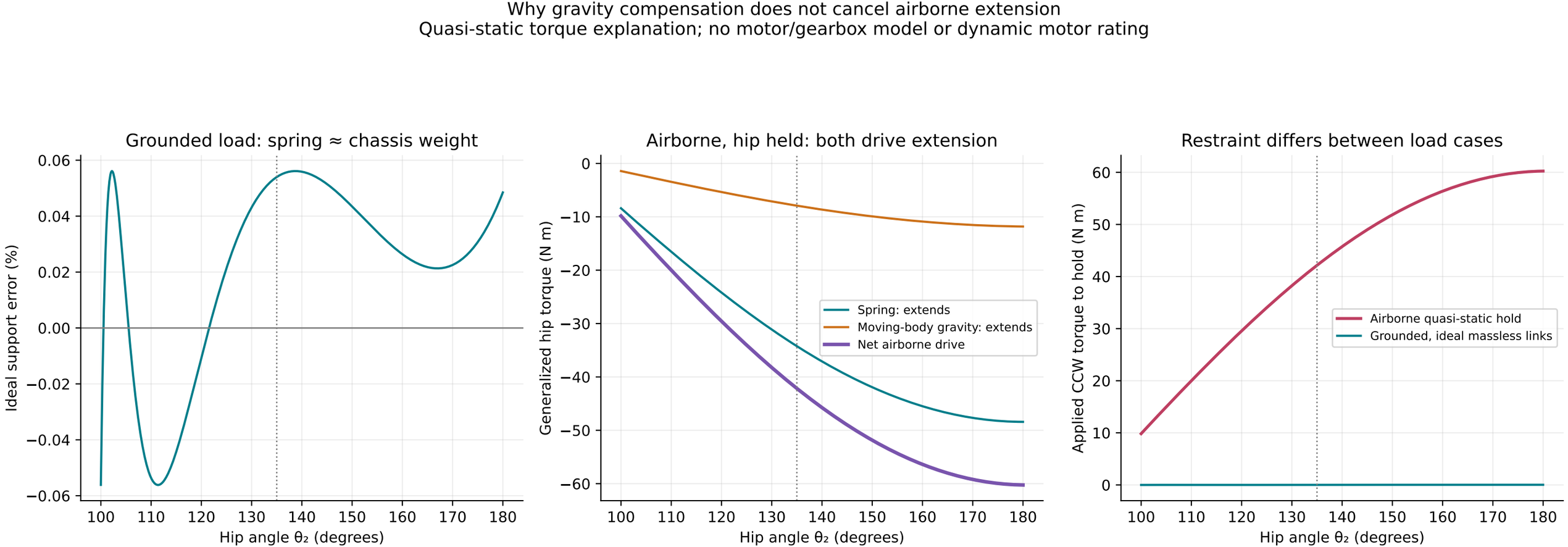
Why gravity compensation does not cancel airborne extension
Quasi-static torque explanation; no motor/gearbox model or dynamic motor rating
Grounded load: spring ≈ chassis weight
Ideal support error (%)
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
100 110 120 130 140 150 160 170 180
Hip angle θ₂ (degrees)
Airborne, hip held: both drive extension
Generalized hip torque (N m)
0
−10
−20
−30
−40
−50
−60
Spring: extends
Moving-body gravity: extends
Net airborne drive
100 110 120 130 140 150 160 170 180
Hip angle θ₂ (degrees)
Restraint differs between load cases
Applied CCW torque to hold (N m)
60
50
40
30
20
10
0
Airborne quasi-static hold
Grounded, ideal massless links
100 110 120 130 140 150 160 170 180
Hip angle θ₂ (degrees)